

Breach Analysis Two

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Reporting

When reporting technical concerns to non-technical senior management there are several things that can make it more effective. The first thing you should try to do to convey your message is translate your message into business impact. If you translate how the technical impact affects business goals, such as budget, timeline, or customer experience it puts things in a more measurable terms that they can relate. Another thing you should try to do is eliminate tech jargon. Saying acronyms for complex terms can make your topic hard to follow to the layman and can lead to frustration or them not understanding the full ramifications that you are trying to express. The third thing that you should do is be concise. The more to the point your presentation is the easier it will be to follow and understand the risk involved.

Sub-Controls

The two additional CIS controls that I would implement in this scenario are 12.6 Deploy Network based IDS sensors and 16.9 Disable Dormant Accounts. Deploying network-based IDS sensors look for unusual attacks or compromises of the system at the boundaries of the network. In this scenario the breach came from an unopened port that was being used for client machines to communicate. A network boundary sensor could have shown that the port was being used for the breach as soon as it was accessed. It could also have shown that the port was being used by client machines communicating via that port. The other CIS control is disabling dormant accounts. By getting rid of accounts that haven't been accessed at a certain time due to employees leaving or accounts not being needed, it migrates the risk that a would-be attacker can use to access portions of the system that aren't being monitored regularly.

Two-factor Authentication

Using RSA encryption is a good way to provide a second trusted credential when logging into a system. This is done using asymmetric security methods of having public and private keys to send data. When logging in, a person uses a digital signature to verify their identity. This is similar to two-factor authentication. It adds another layer onto the logging in process to authenticate the person's credentials. All of this makes it harder for an attacker to use stolen credentials to perform a breach into the system.

VPN Encryption

VPN software uses a combination of asymmetric and symmetric encryption to allow for secure data transfer. Asymmetric encryption is often used to send symmetric keys from one computer to another. (*Everything to Know about VPN Encryption*, 2022) Then after the keys have been transferred the system switches over to symmetric encryption to allow for a faster data transfer. The common type of symmetric encryption used is AES-256 which is an industry standard for security.

References:

Everything to Know About VPN Encryption. (2022, September 9). All about Cookies.

<https://allaboutcookies.org/everything-to-know-about-vpn-encryption>

Microsoft. (2023). *What is two-factor authentication (2FA)? | Microsoft Security.*

Www.microsoft.com; Microsoft. <https://www.microsoft.com/en-us/security/business/security-101/what-is-two-factor-authentication-2fa>